

# Smart Document Collaboration Platform

## Project Overview

Develop a modern web-based **Smart Document Collaboration Platform** where users can create, edit, share, comment on, and manage documents in real time.

The platform should combine **document collaboration, version control, permissions, and intelligent search** into one system.

## Objective

Build a web application similar in concept to a lightweight combination of **Google Docs + Notion + document management**, but with your own implementation and UI.

## Core Requirements

### 1. Authentication

Implement:

- Sign Up
- Login
- Logout
- Forgot Password
- Email Verification
- Session Management

## 2. Workspace

Users should be able to create:

- Workspaces
- Folders
- Documents
- Teams

Each workspace should have its own members and permissions.

## 3. Document Editor

Develop a rich-text editor supporting:

- Headings
- Paragraphs
- Bold / Italic / Underline
- Lists
- Links
- Images
- Tables
- Code blocks
- Quotes
- File attachments

Documents should be automatically saved.

## 4. Real-Time Collaboration

Multiple users should be able to edit the same document simultaneously.

Display:

- Active users
- User cursor/presence

- Real-time changes
- Online/offline status

## 5. Comments

Users should be able to:

- Add comments
- Reply to comments
- Mention users
- Resolve comments
- Delete comments

Example:

@Ali Please update this section.

## 6. Version History

Every major document change should be stored.

Users should be able to:

- View previous versions
- See who made changes
- See when changes were made
- Restore an old version

## 7. Sharing & Permissions

Support:

- Private
- Workspace-only

- Anyone with link

Permission levels:

- Owner
- Editor
- Commenter
- Viewer

## 8. Search

Implement global search for:

- Documents
- Folders
- Users
- Workspace content

Search should support partial matches and filters.

## 9. Notifications

Notify users when:

- Someone mentions them
- A document is shared
- Someone comments
- Someone replies to their comment
- Permission changes
- A document is updated

## 10. Dashboard

Display:

- Recent documents
- My documents
- Shared with me
- Favorites
- Recent activity
- Workspaces

## 11. File Management

Users should be able to:

- Upload files
- Download files
- Rename files
- Delete files
- Move files
- Organize folders

Support common formats such as:

- PDF
- DOCX
- XLSX
- PNG
- JPG

## Backend Requirements

Develop APIs for:

- Authentication
- Users
- Workspaces
- Documents

- Folders
- Comments
- Permissions
- Version History
- File Uploads
- Notifications
- Search

## Database Requirements

Design suitable relationships for:

- Users
- Workspaces
- Workspace Members
- Documents
- Folders
- Comments
- Document Versions
- Permissions
- Notifications
- Files
- Activity Logs

## Technical Requirements

- Responsive design
- Clean architecture
- Reusable components
- REST APIs
- WebSockets for real-time functionality
- Secure authentication

- Role-based access control
- Input validation
- Error handling
- Pagination
- Search optimization
- Proper Git workflow
- Environment variables
- API documentation

## Recommended Stack

### Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS

### Backend

- Node.js / NestJS

or

- Python / FastAPI

### Database

- PostgreSQL

### Real-Time

- WebSockets / Socket.IO

## Storage

- AWS S3 / Cloudflare R2

## Deliverables

- Complete source code
- Functional web application
- Backend APIs
- Database schema / ERD
- API documentation
- README
- Setup instructions
- Deployment
- Demo video

## Deadline

1 Week — 4:00 PM

## Important

A basic CRUD document editor will **not** be considered complete. The key challenge is implementing **real-time collaboration, permissions, comments, version history, and a reliable document workflow.**